

OBJECTIVE

Computer Vision Engineer with 1.5+ years of experience in Research and Development of novel computer vision algorithms and deployment of intelligent solutions, aiming to pursue an MSc in Robotics and AI Systems.

ACADEMIC QUALIFICATION

All India Secondary School Examination, KC International School, Jammu **2016**
CGPA: 10/10

All India Senior School Certificate, Mahavir International School, Jammu **2018**
Percentage: 91.2%

Bachelor of Technology from Indian Institute of Technology (IIT), Bombay **July 2019 - August 2023**
Minor in Artificial Intelligence and Data Science; CGPA: 7.97/10

RESEARCH EXPERIENCE

Computer Vision Research Group | IIT Bombay **April 2022 - May 2023**
Team Size: 10 **Role: Researcher**

- Proposed a novel open-world self-supervised learning framework using Vision Transformers and contrastive learning to address data imbalance and novel class discovery.
- Achieved 84%+ accuracy on unlabeled data and 90%+ on labeled data, demonstrating robustness in low-annotation settings.
- Investigated story visualization using generative models, adapting GANs and diffusion models for coherent visual narrative generation.
- Applied resource-efficient fine-tuning techniques such as prompt-tuning to reduce training cost and accelerate task adaptation.

ACADEMIC PROJECTS

Self-Driving Car Engineer Nanodegree | Udacity **March 2025 - August 2025**

- Built a full autonomous driving software stack, validated in CARLA, covering perception, localization, planning, and control.
- Implemented 3D object detection and multi-object tracking using LiDAR-camera sensor fusion.
- Developed scan-matching-based localization and a motion planner with PID control for stable trajectory execution.

Self-Supervised Speech Synthesis with Visual Input | Automatic Speech Recognition **Spring 2023**

- Implemented AV-HuBERT for self-supervised audio-visual speech enhancement.
- Designed a clean inference pipeline with LoRA-based optimization, reducing inference latency while maintaining performance.

Extended Mask R-CNN | Introduction to Machine Learning **Spring 2023**

- Built a two-stage pipeline for multi-label classification of seven yoga poses on a custom dataset.
- Used Mask R-CNN (COCO-pretrained) for human instance segmentation, followed by a custom neural classifier.
- Achieved 95%+ test accuracy on pose classification.

GPU-accelerated Multi-Agent Trajectories | Optimization in ML **Spring 2022**

- Addressed collision-free trajectory generation for multiple agents under strict constraints.
- Reformulated the problem into a convex quadratic program and solved it efficiently.
- Parallelised the pipeline using JAX, reducing inference time by 57%.

Enhanced Super Resolution GAN | Deep Learning for Remote Sensing **August 2021**

- Developed a GAN capable of $8\times$ image super-resolution for satellite imagery.
- Implemented Residual-in-Residual Dense Blocks to improve generator stability and output quality.
- Integrated perceptual loss and network interpolation, improving perceptual index scores.
- Benchmarked against SRGAN, RCAN, and EDSR, demonstrating competitive performance.

WORK EXPERIENCE

RNT Health Insights, Computer Vision Engineer, Chandigarh, India

September 2025 - Present

- Developing real-time computer vision systems for early gastric cancer detection.
- Building GPU-accelerated edge deployment pipelines achieving sub-20 ms latency.
- Worked with object detection, segmentation, and foundation models, including YOLO, U-Net, and DINO.
- Deployed production systems using NVIDIA DeepStream and Holoscan in Docker-based pipelines.

Nosh Robotics, Computer Vision Engineer, Bengaluru, India

July 2023 - March 2024

- Designed computer vision algorithms for autonomous cooking and real-time monitoring.
- Owned the pipeline end-to-end, from data processing to deployment on ROS.
- Applied knowledge distillation and quantisation to enable real-time inference on embedded hardware.
- Built an LLM-based recipe generation tool to translate culinary inputs into robot-executable scripts.

Bosch, Fixed-Term Associate (Internship), Bengaluru, India

May 2022 - July 2022

- Proposed and evaluated motion planning algorithms for ADAS systems.
- Conducted comparative benchmarking of classical and learning-based planners through literature review.
- Built a ROS-Gazebo benchmarking pipeline to test planners under real-world constraints.
- Designed a *hybrid A-based motion planning algorithm**, achieving an 8× runtime improvement over baseline implementations.

TECHNICAL SKILLS

Languages: Python (5+ years), C++(4+ years), Bash(4+ years), MATLAB (3+ years), CUDA, DeepStream, Holoscan

Machine Learning: Pytorch (6 years), Tensorflow (3+ years), OpenCV (6+ years), JAX, Scikit-Learn

LEADERSHIP ROLES

Avionics and Controls Lead, IIT Bombay Rocket Team

August 2021 - March 2023

- Led the design, development, and testing of onboard avionics and flight software for a student-built rocket.
- Designed a Raspberry Pi-based avionics board using EAGLE, integrating multiple sensors via I²C and implementing SPI-based data logging.
- Conducted ground and powered flight testing of the avionics bay in preparation for the Spaceport America Cup 2023, ensuring system reliability under flight conditions.

Junior Design Engineer, IIT Bombay Racing Team

November 2019 - June 2021

- Fabricated the liquid-cooling system for high-performance motors, balancing thermal efficiency with aerodynamic constraints.
- Designed a passive battery cooling solution using phase-change materials to maintain optimal operating temperatures.
- Contributed to the team securing 1st Place in Engineering Design at FSUK 2020 and a Global Rank of 4.

EXTRACURRICULARS

- Stood first among all the freshmen in the Inter Hostel Music Video Making Competition **2019**
- Bagged 2nd position in Inter-Hostel Music Arcade, performing as lead guitarist **2022**
- Blue Belt Taekwondo athlete **2013**

COMMUNITY INVOLVEMENT

Volunteer, Anmol Cultural Club

June 2024 - December 2024

- Supported NGO initiatives focused on women and child development through grassroots engagement.
- Delivered computer education workshops and career mentoring to underprivileged children, improving digital literacy and awareness of academic pathways.
- Assisted in field operations and program execution, collaborating with staff to ensure timely and effective delivery of community initiatives.

Volunteer, IIT Bombay

2020 - 2023

- Mentored 25+ middle-school students in designing and assembling solar lamps as part of the SoULS (Solar Urja through Localisation for Sustainability) initiative.
- Volunteered for the Powai Lake Mega Clean-Up Campaign organised by Abhyuday, contributing to large-scale environmental awareness and community action.

LANGUAGES KNOWN: English (proficient), Hindi (native), Punjabi (Fluent), Dogri (Fluent), German (Beginner)